

Laboratory Information System (LIS).

(1) this is a hospital workflow management system.

Full life cycle.

Patient Register.

↓
Lab Order Creation.

↓
Sample Collection

↓
Lab processing

↓
Result entry

↓
Clinical Report Generation.

The system ensure.

- Role based access
- workflow integrity
- Result tracking.
- audit safe lifecycle management.

Requirement Stack.

- Django
- SQLite.
- DRF
- Bootstrap 5
- JWT authentication.

Required roles.

Admin - Full system access.

Physician - Register patient + place orders.
+ View reports.

Nurse - Register patient + place orders.

Phlebotomist - Collect samples.

Lab technician - process samples +
Enter results.

Required workflow.

- 1 - Ordered.
- 2 - collected
- 3 - in lab
- 4 - completed.

Required database.

patient - master

test - master - master

assay - master

user - master

user - access - master

Order - transaction.

results.

Required Screens.

- Login
- Dashboards
- patients Registration.
- patient Search
- Lab Order entry.
- phlebotomist work list.
- Technician work list.
- Clinical reports
- Test menu management
- User management.

Required apis.

- jwt auth.
- patient apis
- order
- assay
- result
- report.

Login page.

Purpose.

Authenticate Users and redirect
based on role.

Features.

- Jwt/session authentication.
- role based redirect.
- invalid credential handling.
- Secure login

Why.

Different users different purpose.

Dashboard

Purpose

Central Control panel.

Features

- pending order counts
- Completed reports
- quick navigation.

• Role Specific Sidebar.

Why.

provides workflow visibility.

Patient Registration.

Purpose

Create patient demographic record.

Features.

- MRN validation
- age validation.
- gender Selection
- nationality
- DOB auto handling
- duplicate prevention.

Why

Every laboratory workflow begins with a patient record.

Patient List / Search

Purpose

Search patient quickly

Features

Search by MRN

Search by name

pagination (no did)

Lab report entry

Purpose

Physician / nurse Creates Lab test.

Features

select patient

dynamic assay search

multi test order

add / remove tests

Create Order lines

auto status - Ordered.

phlebotomist worklist

Purpose

Sample collection, management.

Features

• Only ordered status shown

• Collects sample action

• status update

• timestamp tracking

• Collect by tracking.

Why

↑ track chain - of - Custody for Specimens

Technician worklist

Purpose

Lab processing and enter result

Features

• mark in Lab

• Enter assay results

• remove

• Flag • complete workflow.

why

Core diagnostic processing
happen here.

Clinical report view.

Purpose

Doctor view final reports

Features.

- patient search
- Order history
- printable report
- pdf ready layout
- reference ranges
- result flags

Why Important.

Final output for treatment decision.

Test menu management.

Purpose.

Admin Control assay catalog

Features.

Could test categories
Could assays
active/inactive control.

why

Lab frequently update catalogs.

User management.

Purpose.

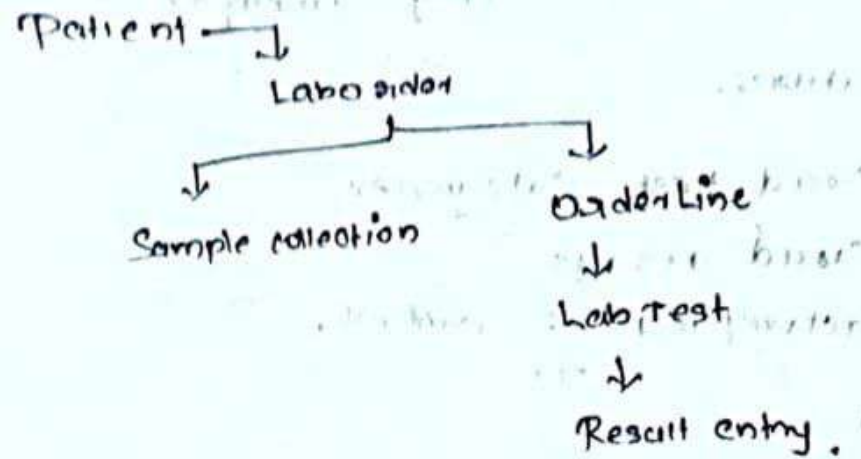
Admin Control Staff access.

Features

- Create users
- assign roles
- manage permissions,

why
Security,

Database Design Explanation.



Introduction.

The project is a Laboratory Information System developed using Django & Django rest framework. It manages the complete lifecycle of Lab testing with strict role-based workflows.

Architecture.

1. Separated the project into multiple Django apps such as accounts, patient, Labtests, orders and results for modularity and maintainance.

Database.

The database follows normalized relational design using foreignkey & one-to-one relationships. OrderLine acts as the bridge between Orders and assays.

Security.

1. Implemented JWT authentication and role based access control to restrict action based on user responsibility.

Workflow.

The most important feature is the forward only Laboratory workflow, Ordered → Collection → in Lab → Completed. Rollback is blocked both at software & business logic.

Api

The system expose Rest apis for
patients, assays, Orders,
reports and workflow transitions.

Frontend

The frontend uses Django
templates with Bootstrap 5
and responsive layouts for
role specific dashboards
and printable reports

• Challenges •

maintaining Strict workflow
integrity across multiple roles

while preventing duplicate Sample
collection.